Dr. HUANG Qirui

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Hiring Manager, AIGC Video Generation
Shopee, Singapore

Dear Hiring Manager,

I am writing to apply for the AIGC Video Generation Algorithm (Leader) role at Shopee. I lead research on world models and generative video at Huawei's Central Research Institute in Singapore, and over the past eight years I have worked across the span this role describes: the distributed systems that train and serve large models, the generative models themselves, and the work of turning research into products.

On the infrastructure side, I led the development of the training and inference engine for Huawei's Pangu models, which runs 10x faster than Microsoft DeepSpeed, and I direct research on large GPU clusters for LLM and multimodal training and inference, from parallelism and mixed precision to long-context serving.

On the model side, I founded and lead a world-model initiative covering long-horizon video generation, real-time interactive world generation and 3D scene generation, including research on diffusion and DiT-based video models. I also led a 1.7B-parameter hybrid Mamba-Transformer mixture-of-experts model for unified multimodal understanding and generation. With only 0.39B active parameters it comes within 0.03 GenEval of Janus-Pro-1B, and at a 1M-token context it prefills 15.8x faster than Qwen3.5-2B with 26% less memory. Long video is above all a long-sequence problem, and this efficiency-first approach to architecture and inference is what I would bring to long-video modeling, temporal consistency, and the distillation and acceleration needed to put video models into production.

I have also built teams from zero. I founded and built Huawei's Singapore research lab from the ground up (the Singapore region has 40 staff), later handing it over to a Chief Expert, and I now lead a five-person core team on generative models and world models. Earlier, at SUTD, I led a 20-person research team on the CISDeM project and delivered a MINDEF-sponsored research programme worth SGD 4.7M. I formulate mid- to long-term technology strategy and investment portfolios for senior leadership, and I have delivered more than ten partner projects worth over USD 10M.

At Shopee's scale, video generation is a product question as much as a research one: product videos, ads and creative content for millions of sellers have to be good, controllable and cheap to generate. I would welcome the chance to build that pipeline end to end, from data and pre-training to alignment, evaluation and deployment, and to grow the team that delivers it.

Thank you for your consideration. I would be glad to discuss how I can contribute.

Sincerely,

Dr. HUANG Qirui